MIL-P-60646B (AR)
24 April 1984
SUPERSEDING
MIL-P-60646A (AR)
10 August 1979

MILITARY SPECIFICATION

PROJECTILE, 105MM, HEAT-T, M456A1
METAL PARTS

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of metal parts for HEAT-T, M456A1 Projectile.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specificatons and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

MIL-C-60536 - Coating, Anodic, Hard, For Aluminum and Aluminum Alloys

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

7548604	-	Pallet (Boxed Type) for Projectile and Projectile Metal Parts
8597604	-	Projectile, 105MM, HEAT-T, M456A1 Metal Parts Assembly
9323970	-	Circumferential Ultrasonic Inspection Standard for Body, Preform
9323971	-	Longitudinal Ultrasonic Inspection Standard for Body, Preform
9323972	-	Envelope Drawing, Ultrasonic Inspection Equipment for Body, Preform

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

ASTM Methods - American Society for Testing and Materials

E8	-	Tension Testing of Metallic Materials
E10	-	Brinell Hardness of Metallic Materials, Test for
B117	-	Salt Spray (Fog) Testing
A322	-	Steel Bars, Alloy, Standard Grades, Spec for
D785	-	Test for Rockwell Hardness of Plastics and Electrical Insulation Materials
D792	-	Test for Specific Gravity and Density of Plastics by Displacement
D2290	-	Test for Apparent Tensile Strength of Ring or Tubular Plastics by Split Disk Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila., PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Heat treatment or stress relief. The heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in each component the mechanical properties specified on the applicable component drawing.

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3.3 Protective coating. Protective coating shall be in accordance with the applicable drawings and specifications. When protective coating is required by component details, finish requirements will apply to all indicated surfaces of the component regardless of contour configuration. Salt spray testing shall be in accordance with the applicable drawings.

3.4 Painting. Painting shall comply with the requirements on the applicable drawings. All paint and marking shall be dry to the touch before testing and packing for shipment.

3.5 Functional

3.5.1 Metal parts security. There shall be no loss, breakup, loosening or deformation of any metal parts component, except the obturating band and seal, in the gunbore or in flight, during any of the proving ground tests conducted.

3.5.2 Target-Impact-Dispersion. The pooled probable error shall be not greater than .20 mils vertical and .24 mils horizontal when tested in accordance with paragraph 4.5.5.2. See 6.5 for method of calculating pooled P.E.

3.6 Process description and controls. The contractor in accordance with the contract provisions shall make available a written description of the specified production process and process controls, determined by the Government as crucial for selected components that are to be used in the production of the first article sample (see paragraph 4.3.5).

3.7 Molding defects. Soundness of material shall be in accordance with the applicable drawing and specification. All molded components shall be free of blisters, cracks, check, inclusions, scratches, voids, and any surface or internal defects.

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the First Article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.9.1 Materials defects. All components shall be free of porosity and excessive flash.

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3.9.2 Metal defects

3.9.2.1 Components. All components, and component cavities shall be free from cracks, splits, cold shuts, inclusions, pipes, porosity, burst, folds, seams, and other metal defects.

3.9.2.2 Component cavities. In addition to 3.9.2.1 all component cavities shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits, craters, and sharp edges.

3.9.3 Burr. No component shall have a burr.

3.9.4 Foreign matter. All parts and assemblies shall be free of dirt, chips, oil, grease, rust, corrosion or other foreign matter prior to inspection and shipping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawings	Quantity
Plug and Disc Assembly	8597489	20
Boom	8597609	20
Fin	8597610	20
Fin and Boom Assembly	8597611	20
Seal	9327359	100
Body	9327358	20
Body and Seal Assembly	9327296	20
Obturator	9327293	100
Locking Ring	10521640	20
Fuze, Lock Plug	8597614	20
Projectile Metal Parts (Prior to Paint)	8597604	20
Projectile Metal Parts (After Paint)	8597604	55
Test Panels - See 4.3.4		

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-60646B (AR)

PARAGRAPH	TITLE: Projectile Metal Parts, M456A1		SHEET 1 OF 2		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
I.	Components and Sub-Assemblies				
	1. 8597609 - Boom	20		3.1	4.4.2.1 & 4.3.4
	2. 8597610 - Fin	20		3.1	4.4.2.2 & 4.3.4
	3. 9327293 - Obturator	100		3.1	4.4.2.3 (See Note 1)
	4. 9327359 - Seal	100		3.1	4.4.2.4 (See Note 2)
	5. 9327296 - Body and Seal Assy.	20		3.1	4.4.2.5
	6. 8597611 - Fin and Boom Assy.	20		3.1	4.4.2.6 & 4.3.4
	7. 9327358 - Body	20		3.1	4.4.2.7
	8. 10521640 - Ring, Locking	20		3.1	4.4.2.8
	9. 8597489 - Plug and Disc Assy.	20		3.1	4.4.2.9
	10. 8597614 - Fuze, Lock Plug	20		3.1	4.4.2.14
NOTE 1.	After the dimensional and visual inspection of the 100 samples is completed, subject 60 samples (20 for each temp.) to the ultimate load and extension test. The remaining 40 samples shall be subjected to the hardness test. After the hardness test is completed, subject 20 samples to the specific gravity test and 20 samples to the moisture content and water absorption test.				
NOTE 2.	Same as Note 1 with the exception that the hardness will not be conducted.				

NOTES:

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-60646B

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
	Projectile Metal Parts, M456A1		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
II.	Projectile Metal Parts (Prior to Paint) - 8597604				
	Examination for Defects and Tests	20		3.1	4.4.2.10
III.	Projectile Metal Parts (After Paint) - 8597604				
	1. Examination for Defects and Tests	55		3.1	4.4.2.11 (See Note 3)
	2. Ballistic Tests				
	a. Target-Impact-Dispersion and Security	25	4.4.3.4.3 and 4.4.3.4.4	3.5.1 and 3.5.2	4.5.5.2
	b. Excess Pressure - Security	25	4.4.3.4.3	3.5.1	4.5.5.1

NOTES:

3. After the dimensional and visual inspection of the 55 samples is completed, subject three samples to the salt spray test and two to the adhesion of paint test. The remaining 50 shall be subjected to the ballistic tests.

DRSMC-DA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.3.4 Abrasion resistance

4.3.4.1 Test panels. When the protective coating is being applied to the first article sample of the fin and boom assembly (or components thereof), 4 test panels as specified in paragraph 9.4.3 of MIL-C-60536 shall be processed along with the product, through each step in the coating process. After coating, the panels will be forwarded to a government approved laboratory for testing in accordance with 9.4.3 of MIL-C-60536 to determine compliance with paragraph 8.4.1 of MIL-C-60536.

4.3.4.2 File test. At the same time the test panels provided by 4.3.4.1 are being subjected to the abrasion test at the Government approved laboratory, 2 fin and boom assemblies (or two sets of component parts thereof) from the first article sample shall be subjected to the file test in accordance with paragraph 9.4.4 of MIL-C-60536 to establish correlation (of hardness of coat) between the panels from 4.3.4.1 and the file test of the fin and boom assembly or components thereof. The file test will be performed using the equipment and procedures that will be used in production testing.

4.3.4.3 Correlation. When performing the file test prescribed by 4.3.4.2, the following information shall be obtained.

a. The type and model of the filing machine used.

b. The type of file (roughness) used.

c. The length of the file stroke.

d. The pressure applied to the file.

e. The number of strokes of the file required to expose the basic metal.

The information thus obtained shall be the basis for determining compliance of the production sample hereinafter prescribed, with the requirements of paragraph 8.4.2 MIL-C-60536.

4.3.4.4 Deletion of the file test. The contractor, at his option, may select the wheel abrasive test (instead of the file test) to be used during regular production. In this case paragraphs 4.3.4.2 and 4.3.4.3 do not apply

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4.3.5 Process description and controls. At the same time as submission of the first article samples, the contractor shall make available a written description of the production process and controls being used in the manufacture of the item. Written descriptions shall be submitted on the following component, body, dwg. 9327358. Included as a minimum shall be the intermediate steps in the press operation used in forming the body as well as intermediate process temperatures and their specific limits.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation of MIL-A-48078.

4.4.1.1 Heat treatment lots. Heat treatment lots shall contain bodies from not more than one heat of steel, that has been heat treated in the same heat-treating equipment, at the same temperature, in one unchanged continuous process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Miscellaneous lots. Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stress relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.3 Projectile metal parts assembly lots. When it has been determined that projectile metal parts of a heat treated or miscellaneous lot have complied with the mechanical properties requirement of the applicable drawings, they may be regrouped for assembly into projectile metal parts assembly lots, to which will be assigned a projectile metal parts assembly lot number. The projectile metal parts assembly lots shall consist of projectile metal parts assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. The projectile metal parts assembly lot thus constituted shall be the basis for selection of samples for the functional tests hereinafter prescribed.

4.4.2 Examination. See MIL-A-48078.

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a. Unless otherwise specified in the classification of defects and tests, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 Level II.

b. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

c. Tool control dimensions. Dimensions marked tool control shall be gaged during first article testing and after each 250 items produced on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action shall immediately be instituted by the contractor. All material produced prior to periodic inspection shall be rejected and screened whenever a defective tool control dimension is found.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.1	Boom				8597609
					NEXT HIGHER ASSEMBLY 8597611
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Minor diameter of body thread, max.		100%%	3.1	Gage
102	Pitch diameter of body thread, max.		0.40%	3.1	Gage
103	Major diameter of fin thread, min.		100%	3.1	Gage
104	Pitch diameter of fin thread, min.		0.40%	3.1	Gage
105	Length of effective fin thread, min.		0.40%	3.1	Gage
106	Length of effective body thread, min.		0.40%	3.1	Gage
107	Length of forward basic diameter		0.40%	3.1	Gage
108	Length of rear basic diameter		0.40%	3.1	Gage
109	Distance from front face to inside basic diameter of cavity		0.40%	3.1	Gage
110	Runout of forward outside diameter with pitch diameter of body thread, and the front face		0.40%	3.1	Gage
111	Runout of tapered section of boom with pitch diameter of body thread, and the front face		0.40%	3.1	Gage
112	Runout of taper in cavity with large outside diameter		0.40%	3.1	Gage
113	Runout of rear straight section of boom with pitch diameter of body thread, and the front face		0.40%	3.1	Gage
114	Flatness of body seat surface		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 8597609
4.4.2.1	Boom				NEXT HIGHER ASSEMBLY 8597611
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Body seat gap		0.40%	3.1	Gage (See Dwg. 8597609 Note 7)
116	Diameter of rear straight section of boom		0.40%	3.1	Gage
117	Outside front diameter		0.40%	3.1	Gage
118	Profile of rear radius of crimp groove		0.40%	3.1	Gage
119	Location of crimp groove		0.40%	3.1	Gage
120	Depth of crimp groove		0.40%	3.1	Gage
121	Threads damaged or malformed		0.40%	3.1	Visual (See 6.2)
122	Metal defective		0.40%	3.9.2	Visual (See 6.2)
123	Tension test	4.4.3.1.2.3		3.1	4.5.1.2.2.2
124	Abrasion resistance test (See Note 1)	4.4.3.3		3.3	4.5.4.2
125	Thickness of protective coating (See Note 1)		0.40%	3.3	Gage
MINOR					
201	Diameter of fin thread pilot		0.65%	3.1	Gage
202	Diameter of body thread counterbore		0.65%	3.1	Gage
203	Diameter of body thread counterbore chamfer, max.		0.65%	3.1	Gage
204	Pitch diameter of body thread, min.		0.65%	3.1	Gage
205	Depth of body thread counterbore		0.65%	3.1	Gage
206	Depth of cavity		0.65%	3.1	Gage

NOTES: 1. This test can be performed in the fin and boom assembly inspection (para. 4.4.2.6)

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 8597609
4.4.2.1	Boom				NEXT HIGHER ASSEMBLY 8597611
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Length of body thread, max.		0.65%	3.1	Gage
208	Length of fin thread pilot, max.		0.65%	3.1	Gage
209	Length from rear face to fin seat		0.65%	3.1	Gage
210	Length from front face to fin seat		0.65%	3.1	Gage
211	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual (See 6.2)
212	Protective coating on fin and body seat(s) excessive		0.65%	3.3	Visual
213	Surface finish, improper		0.65%	3.1	Visual (See 6.2)
214	Foreign matter		0.65%	3.9.4	Visual
215	Burr		0.65%	3.9.3	Visual
216	Pitch diameter of fin thread, max.		0.65%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.2	Fin				8597610
					NEXT HIGHER ASSEMBLY 8597611
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Minor diameter of boom thread, max.		100%	3.1	Gage
102	Pitch diameter of boom thread, max.		0.40%	3.1	Gage
103	Minor diameter of tracer thread, max.		0.40%	3.1	Gage
104	Pitch diameter of tracer thread, max.		0.40%	3.1	Gage
105	Length of effective boom thread, min.		0.40%	3.1	Gage
106	Length of effective tracer thread, min.		0.40%	3.1	Gage
107	Position of tracer cavity with minor diameter of tracer thread		0.40%	3.1	Gage
108	Position of fin hub with minor diameter of tracer thread		0.40%	3.1	Gage
109	Position of fin hub with pitch diameter of boom thread and the front face		0.40%	3.1	Gage
110	Total length		0.40%	3.1	Gage
111	Straightness of fin blades		0.40%	3.1	Gage
112	Diameter of fin blades		0.40%	3.1	Gage
113	Thickness of fin blade web		0.40%	3.1	Gage
114	Angle at front of fin		0.40%	3.1	Gage
115	Diameter of fin hub		0.40%	3.1	Gage
116	Thread(s) damaged or malformed		0.40%	3.1	Visual (See 6.2)
117	Metal defective		0.40%	3.9.2	Visual (See 6.2)
118	Tension test	4.4.3.1.2.3		3.1	4.5.1.2.2.1

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	Fin				8597610
					NEXT HIGHER ASSEMBLY
					8597611
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
119	Abrasion resistance test (See Note 1)	4.4.3.3		3.3	4.5.4.2
120	Thickness of protective coating (See Note 1)		0.40%	3.3	Gage
121	Width of fin blades (T Section)		0.40%	3.1	Gage
122	Angle between fin blades		0.40%	3.1	Gage
MINOR					
201	Pitch diameter of tracer thread, min.		0.65%	3.1	Gage
202	Diameter of tracer thread undercut, max.		0.65%	3.1	Gage
203	Diameter of tracer cavity		0.65%	3.1	Gage
204	Depth of tracer thread cavity		0.65%	3.1	Gage
205	Depth from rear face to bottom of tracer thread cavity		0.65%	3.1	Gage
206	Depth of boom thread cavity		0.65%	3.1	Gage
207	Radii or chamfers missing or incorrect		0.65%	3.1	Visual (see 6.2)
208	Surface finish, improper		0.65%	3.1	Visual (see 6.2)
209	Foreign matter		0.65%	3.9.4	Visual
210	Burr		0.65%	3.9.3	Visual
211	Protective coating on front face excessive		0.65%	3.1	Visual
212	Diameter of boom thread counterbore, max.		0.65%	3.1	Gage
213	Pitch diameter of boom thread, min.		0.65%	3.1	Gage

NOTES: 1. This test can be performed in the fin and boom assembly inspection (para. 4.4.2.6)

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Obturator				9327293
					NEXT HIGHER ASSEMBLY
					8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Inside diameter		0.40%	3.1	Gage
102	Total length		0.40%	3.1	Gage
103	Location of the basic diameter		0.40%	3.1	Gage
104	Runout of outside diameter to inside diameter		0.40%	3.1	Gage
105	Ultimate Load				
	a) Hot	Note 1	2.5%	3.1	4.5.6
	b) 70°F	Note 1	2.5%	3.1	4.5.6
	c) Cold	Note 1	2.5%	3.1	4.5.6
106	Extension				
	a) Hot	Note 1	2.5%	3.1	4.5.6
	b) 70°F	Note 1	2.5%	3.1	4.5.6
	c) Cold	Note 1	2.5%	3.1	4.5.6
107	Hardness	Note 1	2.5%	3.1	4.5.8
108	Specific gravity	3		3.1	4.5.9
109	Moisture content and water absorption	5		3.1	4.5.10
110	Arrows missing, illegible or in the wrong direction		0.40%	3.1	Visual
111	Evidence of poor workmanship		0.40%	3.1	Visual
112	Molding defects		0.40%	3.7	Visual
113	Profile of external taper		0.40%	3.1	Gage
114	Chamfer incorrect		0.40%	3.1	Gage

NOTES: 1. A special inspection level 3 (S-3) with a double sampling plan according to MIL-STD-105D shall be used.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9327293
4.4.2.3	Obturator				NEXT HIGHER ASSEMBLY 8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Surface finish, improper		0 65%	3.1	Visual (see 6.2)
202	Excessive flash		0.65%	3.1	Visual
203	Foreign matter		0.65%	3.9.4	Visual
204	Burr		0.65%	3.9.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9327359
4.4.2.4	Seal				NEXT HIGHER ASSEMBLY 9327296
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Inside diameter		0.40%	3.1	Gage
102	Total length		0.40%	3.1	Gage
103	Inside chamfer		0.40%	3.1	Gage
104	Tensile				
	a) Hot	Note 1	2.5%	3.1	4.5.7
	b) 70°F	Note 1	2.5%	3.1	4.5.7
	c) Cold	Note 1	2.5%	3.1	4.5.7
105	Extension				
	a) Hot	Note 1	2.5%	3.1	4.5.7
	b) 70°F	Note 1	2.5%	3.1	4.5.7
	c) Cold	Note 1	2.5%	3.1	4.5.7
106	Specific gravity	3		3.1	4.5.9
107	Moisture content and water absorption	5		3.1	4.5.10
108	Evidence of poor workmanship		0.40%	3.1	Visual
109	Surface finish, improper		0.40%	3.1	Visual (See 6.2)
110	Molding defects		0.40%	3.7	Visual
111	Quality of the etching bath	4.4.3.6		3.1	4.5.3.2
112	Quality of the mechanical abrasion	4.4.3.7		3.1	4.4.3.7
MINOR	None defined				

NOTES: 1. A special inspection level 3 (S-3) with a double sampling plan according to MIL-STD-105D shall be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Body and Seal Assembly				9327296
					NEXT HIGHER ASSEMBLY
					8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Major diameter of seal		0.40%	3.1	Gage
102	Distance from boom seat surface to rear face of seal flange		0.40%	3.1	Gage
103	Presence of gap at seal/body joint		0.40%	3.1	Gage
104	Runout of major diameter of seal to bourrelets		0.40%	3.1	Gage
105	Seal above flush with boom seat		0.40%	3.1	Gage
106	Sealant not present for full 360°		0.40%	3.1	Visual
107	Sealant build-up on surfaces		0.40%	3.1	Visual
108	Torque test	Note 1		3.1	4.5.3.2
109	Seal above flush with body bourrelet		0.40%	3.1	Gage
MINOR					
201	Body seal area not properly cleaned		0.65%	3.1	Visual
202	Body protective coating damaged		0.65%	3.1	Visual
203	Surface finish improper		0.65%	3.1	Visual (See 6.2)
204	Radius incorrect		0.65%	3.1	Gage
205	Seal below flush with boom seat, max.		0.65%	3.1	Gage
206	Seal below flush with body bourrelet, max.		0.65%	3.1	Gage
207	Foreign matter		0.65%	3.9.4	Visual

NOTES: 1. One body and seal assembly shall be tested at the middle and end of each shift at each station.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.6	Fin and Boom Assembly				8597611
					NEXT HIGHER ASSEMBLY
					8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Gap between fin and boom		100%	3.1	Gage
MAJOR					
101	Weight		0.40%	3.1	Approved scale
102	Torque test		0.40%	3.1	4.5.3.1
103	Total length, max.		0.40%	3.1	Gage
104	Angle between fin blades (Dwg. 8597610)		0.40%	3.1	Gage
105	Thickness of protective coating (if applicable)		0.40%	3.3	Gage
106	Protective coating inadequate or damaged		0.40%	3.3	Visual
107	Threads damaged or malformed		0.40%	3.1	Visual (See 6.2)
108	Presence of sealant		0.40%	3.1	Visual
109	Metal defective		0.40%	3.9.2	Visual (See 6.2)
110	Runout of the fin to the pitch diameter of the body thread and the front face		0.40%	3.1	Gage
111	Abrasion resistance test (if applicable)	4.4.3.3		3.3	4.5.4.2

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.6	Fin and Boom Assembly				8597611
					NEXT HIGHER ASSEMBLY
					8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Protective coating on body seat, excessive		0.65%	3.1	Visual
202	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual (see 6.2)
203	Surface finish, improper		0.65%	3.1	Visual (see 6.2)
204	Foreign matter		0.65%	3.9.4	Visual
205	Burr		0.65%	3.9.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER 9327358
4.4.2.7	Body				NEXT HIGHER ASSEMBLY 9327296
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Metal defective		100% & 4.4.3.2	3.9.2	4.5.2/Ultrasonic
2.	Ultrasonic acceptance stamp missing		100%	3.1	Visual
MAJOR					
101	Distance from seal seat to middle basic diameter, min.		100%	3.1	Gage
102	Distance from boom seat to rear basic diameter, min.		100%	3.1	Gage
103	Distance from front face to rear basic diameter		0.40%	3.1	Gage
104	Distance from front face to cone seat		0.40%	3.1	Gage
105	Distance from front face to boom seat		0.40%	3.1	Gage
106	Distance from front face to middle basic diameter		0.40%	3.1	Gage
107	Distance from front face to forward basic diameter		0.40%	3.1	Gage
108	Distance from front face to rear of forward bourrelet		0.40%	3.1	Gage
NOTES:					

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
4.4.2.7	Body				9327358
					NEXT HIGHER ASSEMBLY
					9327296
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	Distance from rear face to body cavity, min.		0.40%	3.1	Gage
110	Length of effective spike thread, min.		0.40%	3.1	Gage
111	Length of rear bourrelet		0.40%	3.1	Gage
112	Diameter of seal seat		0.40%	3.1	Gage
113	Diameter of forward end of charge cavity		0.40%	3.1	Gage
114	Diameter across wire slots		0.40%	3.1	Gage
115	Diameter of fuze cavity		0.40%	3.1	Gage
116	Diameter of boom thread undercut		0.40%	3.1	Gage
117	Diameter of body		0.40%	3.1	Gage
118	Diameter of bourrelets, min.		0.40%	3.1	Gage
119	Position of diameter of slots with fuze cavity		0.40%	3.1	Gage
120	Major diameter of boom thread, min.		100%	3.1	Gage
121	Pitch diameter of boom thread, min.		0.40%	3.1	Gage
122	Minor diameter of spike thread, max.		0.40%	3.1	Gage
123	Pitch diameter of spike thread, max.		0.40%	3.1	Gage
124	Pitch diameter of fuze lock plug thread, max.		0.40%	3.1	Gage
125	Runout of charge cavity with bourrelets		0.40%	3.1	Gage
126	Runout of body with bourrelets		0.40%	3.1	Gage
127	Runout of fuze cavity with bourrelets		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
4.4.2.7	Body				9327358
					NEXT HIGHER ASSEMBLY
					9327296
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
128	Runout of pitch diameter of boom threads with bourrelets, including perpendicularity of rear face of boom seat		0.40%	3.1	Gage
129	Runout of pitch diameter of spike thread to bourrelets, including perpendicularity of front face		0.40%	3.1	Gage
130	Boom seat gap		0.40%	3.1	Gage (See Note 17 Dwg. 9327358)
131	Spike seat gap		0.40%	3.1	Gage (See note 17 Dwg. 9327358)
132	Minor diameter of fuze lock plug thread, max.		0.40%	3.1	Gage
133	Length of fuze lock plug thread, min.		0.40%	3.1	Gage
134	Flatness of boom seat		0.40%	3.1	Gage
135	Radius of boom thread undercut		0.40%	3.1	Gage
136	Surface finish of charge cavity		100%	3.1	Visual (See Note 1)
137	Surface finish on boom seat		0.40%	3.1	Visual (See 6.2)
138	Hardness test		4.4.3.1.1	3.1	4.5.1.1
139	Tension test		4.4.3.1.2	3.1	4.5.1.2.1
140	Threads damaged or malformed		0.40%	3.1	Visual (See 6.2)

NOTES: 1. Also two samples per shift shall be cut open and inspected per paragraphs 4.4.3.5 and 4.5.11.

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PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER 9327358
4.4.2.7	Body				NEXT HIGHER ASSEMBLY 9327296
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Distance from rear face to boom seat		0.65%	3.1	Gage
202	Distance from rear face to fuze seat surface		0.65%	3.1	Gage
203	Width of boom thread undercut		0.65%	3.1	Gage
204	Width of wire slots		0.65%	3.1	Gage
205	Runout of tapered section of charge cavity to bourrelets		0.65%	3.1	Gage
206	Width of seal seat		0.65%	3.1	Gage
207	Protective coating on seal seat		0.65%	3.1	Visual
208	Protective coating inadequate or excessive		0.65%	3.3	Visual
209	Grooves in boom seat		0.65%	3.1	Visual
210	Relief on rear bourrelet for seal (if applicable)		0.65%	3.1	Visual (See 6.2)
211	Radii or chamfers missing or incorrect		0.65%	3.1	Visual (See 6.2)
212	Surface finish. improper		0.65%	3.1	Visual (See 6.2)
213	Foreign matter		0.65%	3.9.4	Visual
214	Burr		0.65%	3.9.3	Visual
215	Pitch diameter of boom thread, max.		0.65%	3.1	Gage
216	Pitch diameter of spike thread, min.		0.65%	3.1	Gage
217	Pitch diameter of fuze lock plug thread, min		0.65%	3.1	Gage
218	Distance from boom seat to seal forward seat		0.65%	3.1	Gage

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.8	Ring, Locking				10521640
					NEXT HIGHER ASSEMBLY
					8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Runout of inside diameter with pitch diameter of thread		0.40%	3.1	Gage
102	Pitch diameter of thread, min.		0.40%	3.1	Gage
103	Major diameter of thread, min.		0.40%	3.1	Gage
104	Diameter, inside		0.40%	3.1	Gage
105	Length, total		0.40%	3.1	Gage
106	Metal defective		0.40%	3.9.2	Visual (See 6.2)
107	Threads damaged or malformed		0.40%	3.1	Visual (See 6.2)
MINOR					
201	Diameter of thread pilot		0.65%	3.1	Gage
202	Length of thread pilot		0.65%	3.1	Gage
203	Width of wrench slots (if applicable)		0.65%	3.1	Gage
204	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual (See 6.2)
205	Surface finish, improper		0.65%	3.1	Visual (See 6.2)
206	Burr		0.65%	3.9.3	Visual
207	Foreign matter		0.65%	3.9.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.8	Ring, Locking				10521640
					NEXT HIGHER ASSEMBLY 8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
208	Pitch diameter of thread, max.		0.65%	3.1	Gage
209	Diameter of wrench holes (if applicable)		0.65%	3.1	Gage
210	Location of wrench slots or holes (as applicable)		0.65%	3.1	Gage
211	Depth of wrench slots or holes (as applicable)		0.65%	3.1	Gage
212.	Protective coating inadequate or excessive		0.65%	3.3	Visual

NOTES:

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MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Plug and Disc Assembly				8597489
					NEXT HIGHER ASSEMBLY 8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Position of inside diameter with major diameter of thread		0.40%	3.1	Gage
102	Pitch diameter of thread, min.		0.40%	3.1	Gage
103	Major diameter of thread, min.		0.40%	3.1	Gage
104	Diameter of hole		0.40%	3.1	Gage
105	Thickness through body		0.40%	3.1	Gage
106	Thickness of disc		0.40%	3.1	Gage
107	Thread(s) damaged or not full		0.40%	3.1	Visual (See 6.2)
108	Improper assembly		0.40%	3.1	Visual
109	Metal defective		0.40%	3.9.2	Visual (See 6.2)
MINOR					
201	Pitch diameter of thread, max.		0.65%	3.1	Gage
202	Presence and location of wrench holes		0.65%	3.1	Visual (See 6.2)
203	Radii or chamfers missing or incorrect		0.65%	3.1	Visual (See 6.2)
204	Surface finish, improper		0.65%	3.1	Visual (See 6.2)
205	Burr		0.65%	3.9.3	Visual
206	Foreign matter		0.65%	3.9.4	Visual
207	Diameter of wrench holes, min.		0.65%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Projectile Metal Parts (Prior to Paint)				8597604
					NEXT HIGHER ASSEMBLY 9331300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Metal defects		100%	3.9.2	Visual (See 6.2)
MAJOR	None defined				
MINOR					
201	Radii or chamfers missing or incorrect		0.65%	3.1	Visual (See 6.2)
202	Marking missing, incorrect, or illegible		0.65%	3.1	Visual
203	Foreign matter		0.65%	3.9.4	Visual
204	Burr(s) present		0.65%	3.9.3	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Projectile Metal Parts (After Pairt)				8597604
					NEXT HIGHER ASSEMBLY
					9331300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Weight Incorrect		0.40%	3.1	Approved scale
102	Diameter of bourrelets, max.		0.40%	3.1	Gage
103	Improper assembly		0.40%	3.1	Visual
104	Boom thread damaged		0.40%	3.1	Visual (See 6.2)
105	Target-Impact-Dispersion ballistic test	10		3.5.2	4.5.5.2
106	Security (Excess pressure) ballistic test	10		3.5.1	4.5.5.1
107	Salt spray test	See Note 1		3.3	See Note 1
108	Adhesion of paint	See Note 1		3.3	See Note 1
MINOR					
201	Protective coating inadequate or damaged		0.65%	3.3	Visual
202	Paint on threads		0.65%	3.1	Visual
203	Burr(s) present		0.65%	3.9.3	Visual
204	Paint not dry		0.65%	3.4	Visual/Tactile
205	Foreign matter		0.65%	3.9.4	Visual
206	Protective cap(s) missing		0.65%	3.1	Visual
207	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES: 1. The salt spray test (use ASTM B117 to perform this test) and the adhesion of paint test shall be conducted in accordance with TT-C-490. The actual painted bodies shall be used in these tests. The sampling plan and the rejection criteria is defined in TT-C-490.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548604
4.4.2.12	Pallet				NEXT HIGHER ASSEMBLY 9331300
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Strapping missing, loose or broken		0.65%	3.1	Visual and Manual
202	Pallet damaged (unserviceable)		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual
204	Marking missing, incorrect, or illegible		0.65%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Carton (after sealing)				N/A
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Contents loose		0.65%	3.1	Manual
202	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
203	Carton not completely sealed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60646B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.14	Fuze, Lock Plug			8597614
				NEXT HIGHER ASSEMBLY
				8597604
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Pitch diameter of thread, min.		0.40%	3.1	Gage
102	Major diameter of thread, min.		0.40%	3.1	Gage
103	Thickness		0.40%	3.1	Gage
104	Thread damaged or malformed		0.40%	3.1	Visual (See 6.2)
105	Metal defective		0.40%	3.9.2.1	Visual (See 6.2)
MINOR					
201	Pitch diameter of thread, max.		0.65%	3.1	Gage
202	Depth of cavity		0.65%	3.1	Gage
203	Presence and location of wrench hole(s)		0.65%	3.1	Visual
204	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual (See 6.2)
205	Surface finish, improper		0.65%	3.1	Visual (See 6.2)
206	Foreign matter		0.65%	3.9.4	Visual
207	Burr		0.65%	3.9.3	Visual

NOTES:

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4.4.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.5.

4.4.3.1 Mechanical properties

4.4.3.1.1 Hardness

4.4.3.1.1.1 Heat treatment lots. In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined in 4.4.1.1) shall be subjected to a hardness test with a Brinell hardness tester, using a 5 second minimum dwell or by any other method of hardness testing approved by the procuring activity. In any event, details regarding the detailed test and calibrating procedures shall be submitted to the technical agency for approval.

4.4.3.1.1.2 Miscellaneous lots. Miscellaneous heat treatment lots of bodies shall be hardness tested in accordance with 4.4.3.1.1.1.

4.4.3.1.2 Tension test

4.4.3.1.2.1 Heat treatment lots. Two bodies selected from each heat treatment lot (after all units in the lot have been subjected to the heat treatment process - see paragraph 4.4.1.1) shall be subjected to the tension test to determine compliance with the mechanical properties requirements of the applicable drawing and specification. If the size of the heat treatment lot is greater than 2500, two sample bodies shall be selected from each 2500 or fraction thereof. One body selected shall be of the hardest value and the other body of the softest value as determined by 4.4.3.1.1. Failure of any tension test specimen to comply with the yield strength or elongation requirements shall be cause for rejection of the lot. On tangentially transverse specimens the elongation shall be recorded for information only.

As an integral part of his process control, the contractor may establish a hardness range which will yield acceptable mechanical properties to screen heat treatment lots prior to submission to the tension test inspection requirement. Such a hardness range must be based on a demonstrated correlation between hardness and tensile properties. Data and methods used shall be submitted to the technical agency for approval. Bodies which fall outside of this hardness range can be reprocessed as part of the continuous process lot. Also the bodies which fall outside of the hardness range can be grouped in a miscellaneous lot and reprocessed in accordance with the miscellaneous heat treatment lot provisions.

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4.4.3.1.2.2 Miscellaneous. Four sample bodies shall be selected from each miscellaneous heat treatment lot (see 4.4.1.2) and subjected to the tension test to determine compliance with the mechanical properties requirements. If the size of the miscellaneous lot is greater than 2500, four samples (two of the hardest value and two of the softest value as determined by 4.4.3.1.1) shall be selected from each 2500 or fraction thereof. Failure of one or more tension test specimens to comply with the yield strength or elongation requirements shall be cause for rejection of the miscellaneous lot. On tangentially transverse specimens the elongations shall be for information only. See the process control provisions of paragraph 4.4.3.1.2.1.

4.4.3.1.2.3 Finished aluminum components. The sample size for the tension test shall be on the basis of two tension test specimens for each 1200 components, or fraction thereof, representing forging or extrusions of the same alloy, temper, and design. Failure of any specimens to comply with the applicable requirements shall be cause for rejection of the lot.

4.4.3.2 Ultrasonic inspection. Each body, after final machining except for threads and wire slots, shall be inspected for projectile integrity and metal defects using, as reject standards, drawings 9323970 and 9323971, modified and approved, as applicable, by the technical agency (DRSMC-QAT-A(D) and DRSMC-LCU-CV(D)), and equipment satisfying the requirements of drawing 9323972 and 4.4.4. Deviation from the standards are permitted, provided the changes are approved by the technical agency (DRSMC-QAT-A(D) and DRSMC-LCU-CV(D)). All bodies accepted by the ultrasonic equipment shall be stamped with a "U" on the projectile body. Bodies rejected by the equipment shall be stamped with an "R" on the projectile body and the rejected bodies subsequently removed. The quantities of projectile bodies inspected and rejected shall be counted and logged for determination of the reject rate.

4.4.3.2.1 Ultrasonic equipment calibration. The ultrasonic inspection equipment, specified by drawing 9323972, shall be calibrated, as a minimum, at the beginning, middle, and end of an 8 hour work shift. If for any reason the reject standards are not rejected by any of the channels the contractor must re-calibrate the equipment and re-inspect all the accepted rounds ultrasonically inspected since the last acceptable calibration.

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4.4.3.2.2 Rejection rate - ultrasonic equipment. When the ultrasonic inspection equipment rejection rate exceeds 5.0 percent for the period between calibrations, the contractor shall check equipment calibration. If the calibration is acceptable all projectiles previously accepted by the equipment during this period shall be re-inspected. During re-inspection if the reject rate exceeds 1.0 percent all projectiles scheduled for re-inspection or accepted by re-inspection shall be placed aside until the cause for the excessive reject rate is determined. When the cause is determined, disposition of the held projectiles shall be at the discretion of the contracting officer. At any time if the contractor proves that an excessive reject rate is equipment related he may apply to the contracting officer for re-inspection of those rejected projectiles found during this time.

4.4.3.3 Abrasion resistance test

4.4.3.3.1 Fin and boom assembly - file abrasive test. One fin and boom assembly (or set of components thereof) from each eight hours production or fraction thereof from the protective coating process shall be subjected to this test to determine compliance with paragraph 9.4.4 of MIL-C-60536, except that the information obtained during the file test in 4.3.4.2 shall be utilized as indicated in 4.3.4.3.. Failure of the assembly or components to comply with the applicable requirements shall be cause for rejection of the product processed since the last acceptable test was performed.

4.4.3.3.2 Test panels - wheel abrasive test. If the contractor decides to use the wheel abrasive test during regular production (see 4.3.4.4), two test panels as specified in paragraph 9.4.3 of MIL-C-60536 shall be processed along with the product from each eight hours production. These test panels shall be subjected to a test in accordance with paragraph 9.4.3 of MIL-C-60536 to determine compliance with paragraph 8.4.1 of MIL-C-60536. Failure of any test panel to comply with the applicable requirement shall be cause for rejection of the product processed since the last acceptable test was performed.

4.4.3.4 Functional. The following proving ground tests will be performed by the government.

4.4.3.4.1 First article sample. The quantities listed in Table II shall be ballistically tested in the phases indicated.

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TABLE II

Phase	Temp.	Tube Wear	Quantity	Requirement
Target-Impact-Dispersion	70 ± 2.5°F (21 ± 1.5°C)	New	25	Target-impact-dispersion & metal parts security. Muzzle velocity, pressure, obturator discard, and spin rate for information only.
Excess Pressure	70 ± 2.5°F (21 ± 1.5°C)	New	25	Metal parts security. All others for information only.

4.4.3.4.2 Production lot sample. The quantities listed in Table III shall be ballistically tested in the phases indicated.

TABLE III

Phase	Temp.	Tube Wear	Quantity	Requirement
Target-Impact-Dispersion	70 ± 2.5°F (21 ± 1.5°C)	New	10	Target-impact-dispersion & metal parts security. Muzzle Velocity, pressure, obturator discard for information only.
Excess Pressure	70 ± 2.5°F (21 ± 1.5°C)	New	10	Metal parts security. All others for information only.

4.4.3.4.3 Security. Failure of any projectile assembly to comply with 3.5.1 shall be cause for rejection of the first article sample or the production lot during the performance of any ballistic test.

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4.4.3.4.4 Target-impact-dispersion. Samples provided in accordance with Tables II and III will be subjected to this test. If the pooled Probable Error (PE) of the sample, calculated in accordance with 6.5, multiplied by the appropriate factor of Table IV (see 6.11) exceeds the requirements of paragraph 3.5.2, the lot shall be rejected.

TABLE IV

Sample Size	Degrees of Freedom	Factor
25 - Test Rounds	20	.80
24 - (one target miss)	19	.80
15 - Calibration Rounds	12	.76
10 - Test or Calibration Rounds	8	.72
9 - (one target miss)	7	.71

4.4.3.5 Interior body surface inspection. In order to determine conformance of the interior surface of the body with drawing and specification requirements, two (2) samples shall be inspected from each shift's production as per 4.5.11. The sample shall be a body which was formed by a punch which has been used for a minimum of fifty (50) operations. The samples shall be taken one from the middle of the work shift and one from the end of the work shift. An additional sample shall be taken, if any change to the production process is made, directly after the change is made. Failure of any sample to comply with the requirements of the applicable drawings and specifications referenced thereon shall be cause for rejection of the product produced subsequent to the last inspection.

4.4.3.6 Adhesion testing. The quality of the etching bath shall be monitored by assembling a seal, etched in the bath that is to be tested, to a body and torque testing the assembly. The sample size shall be two assemblies per occasion. The etch solution shall be tested before and after each change to the bath and no less than once every eight hours of production. The samples shall be tested in accordance with paragraph 4.5.3.2. Failure of any sample to comply with the minimum torque requirement of the applicable drawing shall be cause for rejection of all seals etched with that bath or etched subsequent to the last test, whichever is applicable (see 6.10).

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4.4.3.7 Mechanical abrasion (if applicable). The quality of the mechanical abrasion of the seals shall be monitored by taking five (5) seals at the middle and end of each shift and inspecting for the following characteristics:

a. Presence of surface sheen
b. Increase in inside diameter
c. Average surface finish increase (to be measured in at least 3 places, 120° apart).

Failure of any sample to comply with the requirements of the applicable drawing shall be cause for rejection of all seals mechanically abraded since the last acceptable test.

4.4.3.8. Seal assembly test. At the middle and end of each shift one body and seal assembly per station shall be tested in accordance with paragraph 4.5.3.2. Failure of any sample shall be cause for rejection of all assemblies assembled since the last acceptable test.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment design in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.7 herein.

4.5 Methods of inspection

4.5.1 Mechanical properties. The minimum mechanical properties requirements shall apply throughout the component unless otherwise designated. The number of specimens and their location are for the convenience of the contractor and government and shall not restrain the contractor or government from testing additional specimens at other locations to determine that properties of the components conform to drawing requirements.

4.5.1.1 Hardness. The Brinell hardness test shall be made on the rear face of the body. Readings shall be taken at one location. Prior to performing the test, the area in which the reading is to be taken shall be cleaned of scale or other foreign matter. The test shall be performed using equipment and procedure prescribed by ASTM E10.

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4.5.1.2 Tension test

4.5.1.2.1 Body. From the samples provided by 4.4.3.1.2.1, or 4.4.3.1.2.2, two (2) tension test specimens conforming as closely as possible to the largest obtainable standard round specimen as prescribed by ASTM E8 shall be taken longitudinally from opposite sides of the body wall or any other location that may be indicated by the government. Also two (2) tension test specimens conforming as closely as possible to a standard round specimen shall be taken in a tangentially transverse direction on opposite sides of the body at the midpoint of the seal seat area. Yield strength shall be determined by the offset method using a limiting set of 0.1 percent offset. The tension test shall be performed utilizing the equipment and procedures prescribed by ASTM E8.

4.5.1.2.2 Aluminum components

4.5.1.2.2.1 Fin. From each sample supplied in accordance with 4.4.3.1.2.3, one tension test specimen conforming to the largest obtainable round specimen as prescribed by ASTM E8, shall be taken from the small diameter (hub) of each sample. The sample shall be tested using the criteria for performing this test using the offset method as prescribed by ASTM E8. The limiting set shall be 0.2 percent.

4.5.1.2.2.2 Boom. From each sample supplied in accordance with 4.4.3.1.2.3, one tension test specimen conforming to the largest obtainable round specimen as prescribed by ASTM E8, shall be taken at a location that is at the convenience of the contractor. The sample shall be tested using the criteria for performing this test using the offset method as prescribed by ASTM E8. The limiting set shall be 0.2 percent.

4.5.2 Ultrasonic inspection. The equipment and procedures used for performing the ultrasonic flaw inspection of tempered bodies of 4.4.3.2 will be as prescribed by Drawing 9323972. The contractor shall prepare a detailed written standard operating procedure to include the testing and calibration procedures which must be reviewed and approved by the responsible technical agency.

4.5.2.1 Ultrasonic equipment - operators qualification. The operators of the contractors equipment for ultrasonic inspections required by this specification shall be certified and qualified "Level I". Also any set-up, calibration, or interpretation personnel shall be certified and qualified "Level II" The qualification and certification criteria shall be as prescribed by MIL-STD-410 and other applicable specifications.

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4.5.3 Torque test

4.5.3.1 Fin and boom assembly. The fin and boom assemblies shall be mounted in an approved holding device in such a manner that the boom is prevented from turning. The required disassembly torque shall be applied to the fin gradually in the direction required for disassembly. Any perceptible movement of the fin shall be cause for rejection of the fin and boom assembly. Testing shall be conducted after thread sealant is fully cured.

4.5.3.2 Body and seal assembly. The body and seal assembly shall be mounted in such a manner that the body is prevented from moving. A line is then scribed across both the body and seal in such a manner that movement can be easily observed. The torque fixture shall be placed over the unmachined seal and radial holes drilled into the seal for the four holding pins. Holes shall be drilled to a depth to prevent seal tearing but not into body material. The holding pins shall be inserted into the holes and a static torque applied to the seal. No torque shall be applied to the seal until the adhesive is fully cured. The test fixture shall be in accordance with Figure II or its equivalent.

4.5.4 Protective coating

4.5.4.1 Body. The procedures for testing the cleaning of the body assembly and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification.

4.5.4.2 Abrasion resistance. The sample provided by 4.4.3.3.1 shall be tested using a filing machine of a type and model that was used in establishing correlation as indicated in 4.3.4.2. The samples shall be tested using the same type (roughness) file, the same length of stroke, the same number of strokes and the same file pressure as was used on the first article sample. If the contractor decides to do the wheel abrasive test (4.4.3.3.2), the test must be performed in accordance with paragraph 9.4.3 of MIL-C-60536.

4.5.5 Functional tests. All ballistic tests shall be conducted at a Government proving ground in accordance with the applicable approved Acceptance Test Procedure. The sample cartridges shall be fired from a M60 tank series and observations made for conformance to all applicable requirements. Photographic equipment will be set up near the muzzle in such a manner as to record security of projectile metal parts. When ballistically testing the projectile metal parts, reference/calibration components shall be used as a minimum in the target-impact-dispersion phase.

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4.5.5.1 Excess pressure test. The metal parts assemblies supplied for the excess pressure phase, in accordance with 4.4.3.4.1 and 4.4.3.4.2 will be inert loaded and assembled into complete rounds and fired. The propellant charge will be adjusted to obtain a chamber pressure of 70,000 psi (482.6 MPa) to 73,900 psi (509.4 MPa).

4.5.5.2 Target-Impact-Dispersion test.

4.5.5.2.1 Firing sequence. The metal parts assemblies supplied in accordance with this phase of 4.4.3.4.4, inert loaded and assembled into complete rounds, will be fired according to the applicable Acceptance Test Procedure utilizing the service charge. The data generated in the new tube phase will primarily be used for determination of acceptance from a target-impact-dispersion standpoint, and will require the firing of reference/calibration rounds to establish a velocity correction. The firing sequence for the first article sample will be five (5) calibration rounds, fifteen (15) test rounds, five (5) calibration rounds, ten (10) test rounds, five (5) calibration rounds. For the production lot sample the firing sequence will be five (5) calibration rounds, ten (10) test rounds, five (5) calibration rounds. Velocity, spin rate and pressure data shall be recorded during the first article test. For the production lot tests, velocity data shall be recorded for each round fired (for info only, see 6.1). Also for the production lot tests, pressure data shall be collected during the first five lots, then every fifth lot thereafter, for info only.

4.5.5.2.2 Target. Target distance is to be 2000 meters and target size is to be 20-feet X 20-feet (6.1 meters X 6.1 meters) minimum.

4.5.5.2.3 Temperature conditioning. The ammunition shall be conditioned to the appropriate temperature until all propellant is stabilized at a uniform temperature prior to firing. This normally requires a period of 24 hours.

4.5.5.2.4 Aiming. After loading and locking a round in the chamber, the gun tube shall be traversed off target center, dropped down below target center, traversed to the opposite side of target center, elevated to target center elevation, and traversed onto target center. The distance traversed and dropped should be sufficient to remove the backlash from the system. If the sight is above, or beyond target center do not reverse direction of gun tube movement, but repeat the sequence.

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4.5.5.2.5 Spotting the target. Spotter rounds will be used to assure the weapon is zeroed on target center. A minimum of three consecutive target hits within one mil of target center are required. After the target has been spotted for a shot group, every effort should be made to keep the elevation and deflection of the weapon the same for that shot group.

4.5.5.2.6 Firing groups. The test samples shall be fired in five round groups (or occasions) with no less than three minutes but not more than five minutes between consecutive rounds in the same group. If more than a thirty minute lapse occurs between group firings, or if there was an indication of a trend in the previous group's target impacts to move away from the center of the target the weapon will be rezeroed on target center before firing the next group.

4.5.5.2.7 Target impact measurements. Target impacts (including spotters) shall be measured to the nearest one-half inch (1 centimeter) in both the horizontal and vertical distance from target center.

4.5.5.2.8 Wind velocity.-Wind velocity and gusts shall be measured at the firing site prior to the firing of each round in the target-dispersion phase ballistic test. The dispersion portion of the ballistic test shall not be fired if the surface wind velocity (nominal) exceeds 10 knots (18.5 Km/hr) and surface wind gusts exceeds 5 knots (9.25 Km/hr). In addition, peak surface wind velocity (nominal plus gusts) is not to exceed 10 knots (18.5 Km/hr) during the firing of any group during the dispersion phase.

4.5.5.3 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, the results shall be invalidated and additional cartridges, as required, shall be tested. A lot shall be retested whenever it fails to meet the target-impact-dispersion requirements as a result of being affected by the following conditions:

a. The wind velocity exceeds 10 knots (18.5 Km/hr) during the firing of any test group.

b. The cadence between any two consecutive rounds in a test group exceeds 5 minutes.

c. A change was made in the azimuth or elevation of the weapon within a test group.

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d. Both the test samples and the calibration samples fail to meet the target-impact-dispersion requirements.

4.5.5.3.1 Target miss. Projectile lots which had one target miss during the target-impact-dispersion phase of the proving ground test described herein, shall be retested if the pooled PE of the remaining rounds multiplied by the appropriate factor (see 4.4.3.4.4) meets the requirements of 3.5.2. Two or more target misses by the test rounds shall reject the lot. Target miss coordinates will be estimated and recorded for all test phases, along with pertinent flight observations. Results of retests for target-impact-dispersion shall be referred to the contracting officer for disposition.

4.5.6 Obturator, extension and ultimate load. The ultimate load and extension shall be determined in accordance with ASTM D-2290 with the below listed exceptions. The test fixture shall be in accordance with Figure I or its equivalent. Machine loading shall be 0.5 inches per minute $\pm$ 0.1 inches per minute. Sample specimens each shall be tested at -35°F $\pm$ 4°F (-37°C $\pm$ 2°C), 73°F $\pm$ 4°F (22°C $\pm$ 2°C), and 140°F $\pm$ 4°F (60°C $\pm$ 2°C). The ultimate load will be the maximum force recorded during testing and will be independent of cross-sectional area. For elongation the split disc shall be separated until each half is in contact with the specimen without applying a load to the specimen. This distance of separation (initial separation) shall then be measured. The specimen rupture force shall be measured. Split disc separation upon specimen rupture (ultimate separation) shall be measured. Extension shall be determined by subtracting initial separation from ultimate separation. Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot. The specimens used for this test shall be as received obturators.

4.5.7 Seal, extension and tensile. The ultimate tensile strength and extension shall be determined in accordance with ASTM D-2290. The test fixture shall be in accordance with Figure 1 or its equivalent. Sample specimens each shall be tested at -35°F $\pm$ 4°F (-37°C $\pm$ 2°C), 73°F $\pm$ 4°F (22°C $\pm$ 2°C), and 140°F $\pm$ 4°F (60°C $\pm$ 2°C). The split disc shall be separated until each half is in contact with the specimen without applying a load to the specimen. The distance of separation

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(initial separation) shall then be measured. Split disc separation upon specimen rupture (ultimate separation) shall be measured. Extension shall be determined by subtracting initial separation from ultimate separation. Machine loading shall be two (2) inches per minute ± 0.5 inches per minute. Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot. The specimens used for this test shall be as received seals.

4.5.8 Obturator hardness. Rockwell hardness determination shall be made in accordance with ASTM D785, using the R scale with the exceptions that the hardness test shall be performed on the actual machined obturator and the specimens shall be conditioned for a minimum of 24 hours. Four (4) readings 90° apart shall be taken on the outside of the sample (radial direction). Failure of any reading to meet the applicable requirement shall be cause for rejection of the lot.

4.5.9 Seal and obturator, specific gravity. Three (3) seals and three (3) obturators shall be selected from each of the lots of seals and obturators for use in determining the specific gravity. The test shall be performed in accordance with method A1 per ASTM D792. Failure of any sample to meet the applicable requirement shall be cause for rejection of the lot.

4.5.10 Seal and obturator, moisture content and water absorption. The bands shall be wiped clean with an OSHA acceptable solvent, i.e., dichlorotetraflourathane, force air dried, then weighed on an analytical balance. This is the initial weight. The bands are then placed in an oven maintained at 160°F ± 4°F (71°C ± 2°C) for twenty-four (24) hours. Cool in a desiccator and reweigh. Repeat oven conditioning and cooling until constant weight is achieved. This is the conditioned weight. Completely immerse the dried bands in a container of distilled water, maintained at 73°F ± 4°F (22°C ± 2°C) for a period of twenty-four (24) hours minus zero (0) plus one-half (1/2) hour. Bands shall be individually removed from the water, all surface water wiped off with a dry, lint free cloth and weighed immediately on an analytical balance. This gives the wet weight. Percent moisture content shall be calculated as follows:

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$$\frac{\text{Initial weight} - \text{conditioned weight}}{\text{Initial weight}} \times 100 = \%\ \text{moisture content}$$

Percent water absorption shall be calculated as follows:

$$\frac{\text{Wet weight} - \text{conditioned weight}}{\text{Conditioned weight}} \times 100 = \%\ \text{water absorption}$$

Failure of any sample to meet the applicable requirement shall be cause for rejection of the lot.

4.5.11 Interior surface. The samples taken in accordance with 4.4.3.5 shall be cut in half along a center line of the body. Surface finish shall be determined by a visual inspection and at the same time blend of contour shall also be examined.

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level B minimum military pack. The metal parts for projectile, 105MM, HEAT-T, M456A1 shall be packed as follows:

5.1.1.1 Metal parts assembly. Part No. 8597604 shall be packed in accordance with drawing 7548604.

5.1.1.2 Fin and boom assembly. Part No. 8597611 shall be packed in accordance with MIL-STD-1169. Pack sixteen (16) fin and boom assemblies arranged in a 4 X 4 pattern, using interlocking separators.

5.1.1.3 Plug and disc assembly. Part No. 8597489 shall be packed in accordance with MIL-STD-1169. Pack 2500 Plug and Disc Assemblies arranged in a 10 X 10 pattern, using interlocking separators.

5.1.1.4 Obturator. Part No. 9327293 shall be packed in accordance with MIL-STD-1169. Pack 750 obturators arranged in a 5 X 5 pattern, 10 to a cell, in three (3) layers, using interlocking separators.

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5.1.1.5 Locking ring. Part number 10521640 shall be packed in accordance with MIL-STD-1169.

5.1.1.6 Fuze, lockplug. Part number 8597614 shall be packed in accordance with MIL-STD-1169.

5.1.2 Percentage of extra parts packing. Packing for the required percentage of extra parts shall be provided for by furnishing the proper number of prime containers.

5.1.3 Shipping. If the component parts from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated.

5.1.4 Marking. The pallet shall be marked in accordance with drawing 7548604. The carton shall be marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Velocity. Proving ground tests for velocity are not acceptance tests, but the results will be carefully analyzed by US ARDC to determine quality of the ammunition produced.

6.2 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.3 In checking the thickness of protective coating on the fin and base the "Dermitron" by Unit Process Assemblies Inc. 53-15 37th Ave., Woodside, N.Y. 11377, has been proven to be satisfactory.

6.4 Ordering data. See MIL-A-48078.

6.5 Method of calculating pooled PE. The individual PE results of the groups (5 samples in each group) of samples shall be pooled in accordance with the following formula:

$$PE_{pooled} = \sqrt{\frac{\sum\left[(N_i - 1)\ PE_i^2\right]}{\sum (N_i - 1)}}$$

Where: i - Number of groups
N_i - Number of samples in a group (target hits)
PE_i - Probable Error of the group

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6.6 New Tube wear definitions. A new tube shall not have the primary land wear exceed .015 inches. Primary wear is measured in a vertical plane 25.25 inches from the rear face of the tube.

6.7 Submission of inspection equipment designs for approval. See MIL-A48078.

Commander
US Army Armament Research and Development Center
ATTN: DRSMC-QAT-I(D)
Dover, N.J. 07801

6.8 Submission of test data and ammunition data cards. In addition to the normal distribution of records when the projectile is procured by the Army Armament, Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARDC, ATTN: DRSMC-LCU-M(D), DRSMC-LCU-CV(D) and DRSMC-QAT-A(D), Dover, NJ 07801.

6.9 Sulfuric acid - Dicromate solution. The sulfuric acid dicromate solution shall be prepared in accordance with the applicable drawing.

6.10 Seal rework. If any test specimen fails to meet the requirement the rejected seals may be re-etched provided that the total etch time does not exceed the maximum allotted time.

6.11 Probable error factor. The factors shown in Table IV are taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limit (A .05) and (N-1) degrees of freedom.

6.12 Probable error calculations. The probable error of the group is calculated in accordance with the following equation:

$$PE_{mils} = \frac{(.6745)\ (1000)\ (SD)}{(X)\ (Rg)}$$

Where:

SD - Standard deviation
Rg - Range in meters
X - Conversion factor
a) when target hits measured in inches,
X = 39.37
b) When target hits are measured in centimeters,
X = 100.0

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6.13 Standard deviation. The standard deviation of a sample is calculated in accordance with the following equation found on page 1-10 AMCP 706-110:

$$SD = \frac{N \sum X^2 - \left(\sum X\right)^2}{N(N-1)}$$

Where:

N - the number in the sample
X - the individual data point

6.14 Visual standards. Visual standards are to be established at the contractor's plant for compliance with 3.9.2.

6.15 Drawings. Drawings listed in section 2 of this specification under the heading U. S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

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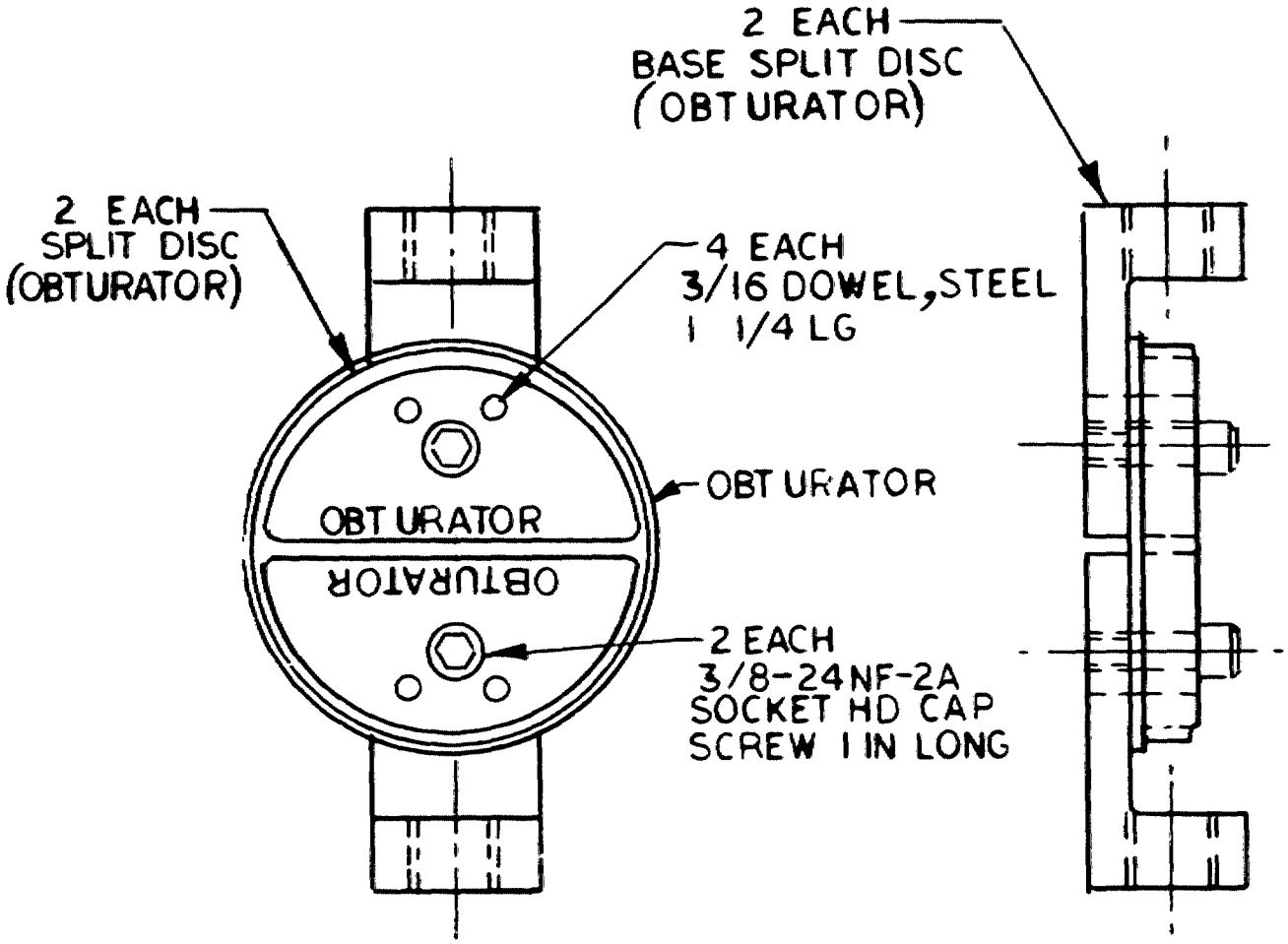
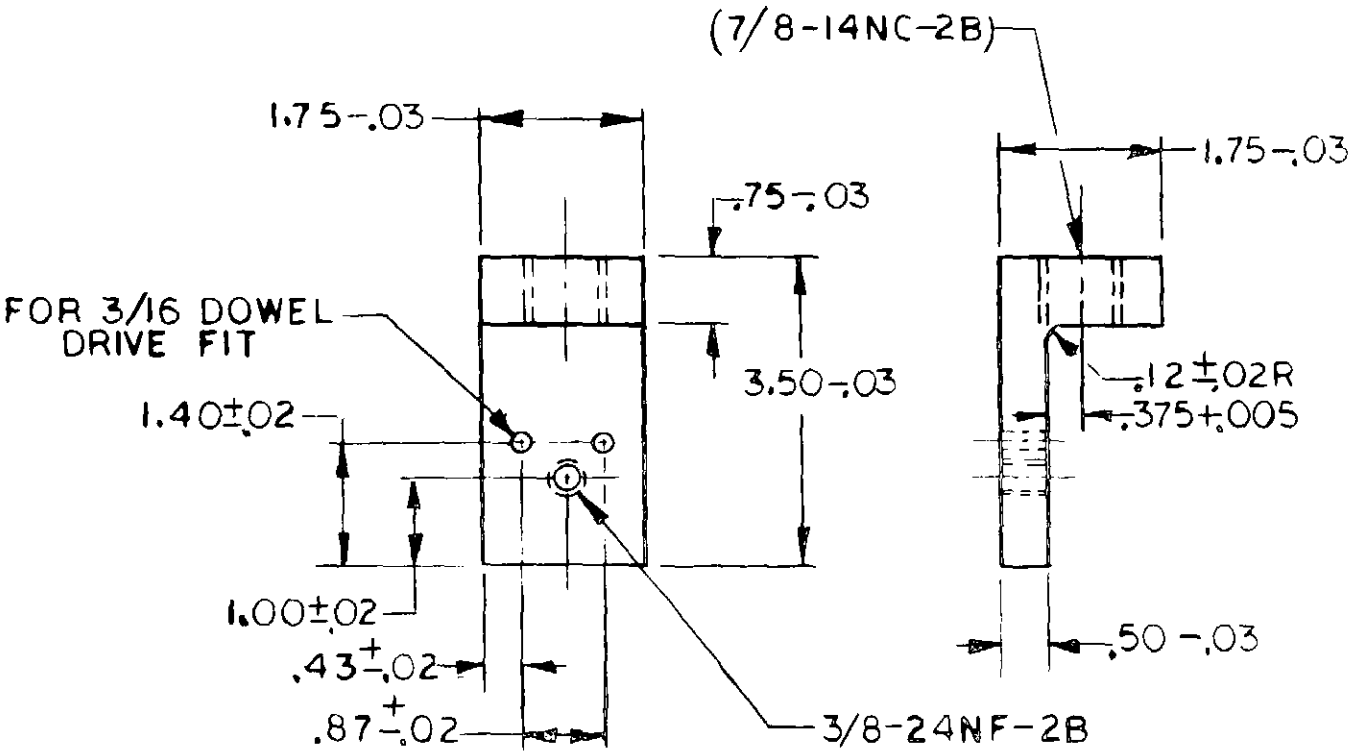


FIGURE 1.A OBTURATOR SPLIT DISC ASSEMBLY

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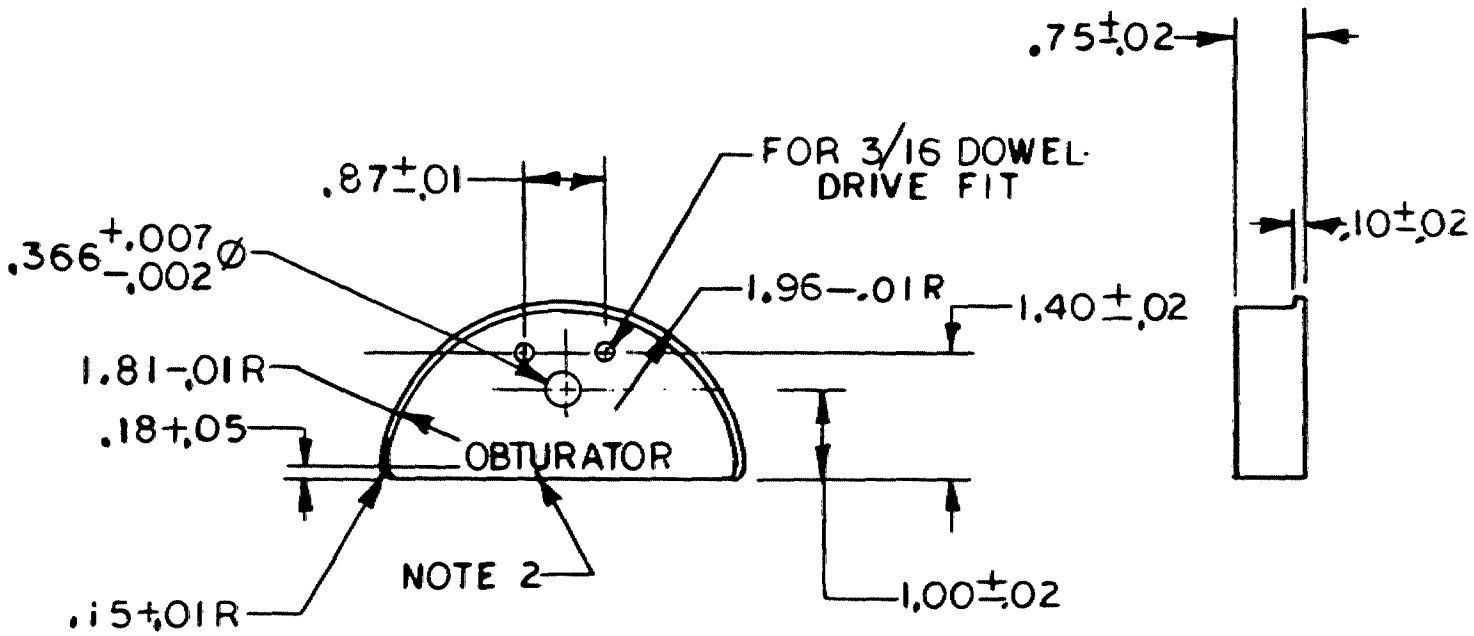


NOTES:-

1- MATERIAL:- STEEL, HOT-ROLLED ALLOY BARS, GRADE 4340, ASTM-A322, 2 REQ'D

FIGURE 1.B BASE SPLIT DISC (OBTURATOR)

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NOTES:—

1—MATERIAL:—STEEL, HOT-ROLLED ALLOY BARS, GRADE 4340, ASTM-A322, 2 REQ'D.

2—STAMP AS SHOWN WITH .25-.06 HIGH LETTERS.

FIGURE 1.C SPLIT DISC (OBTURATOR)

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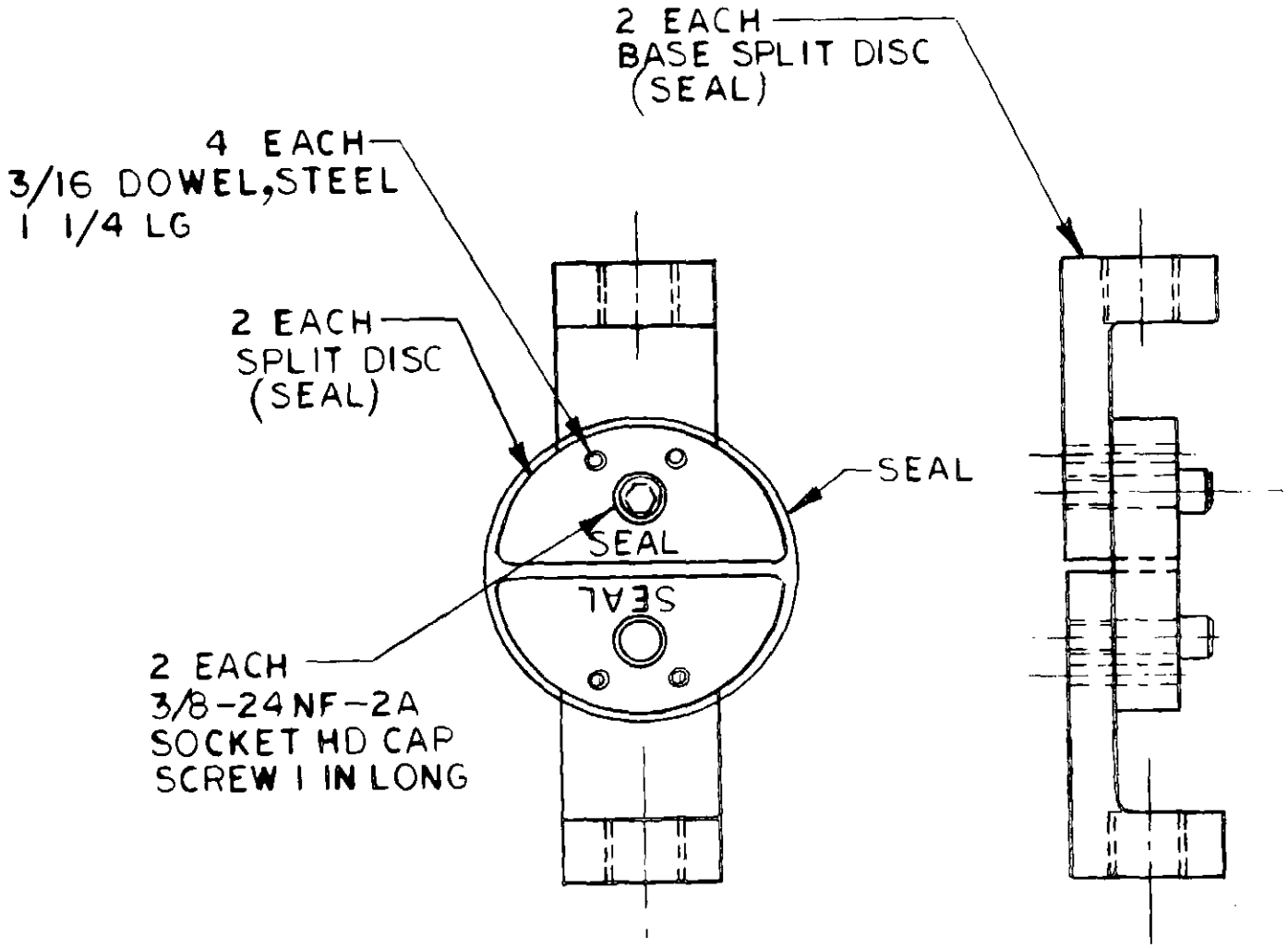
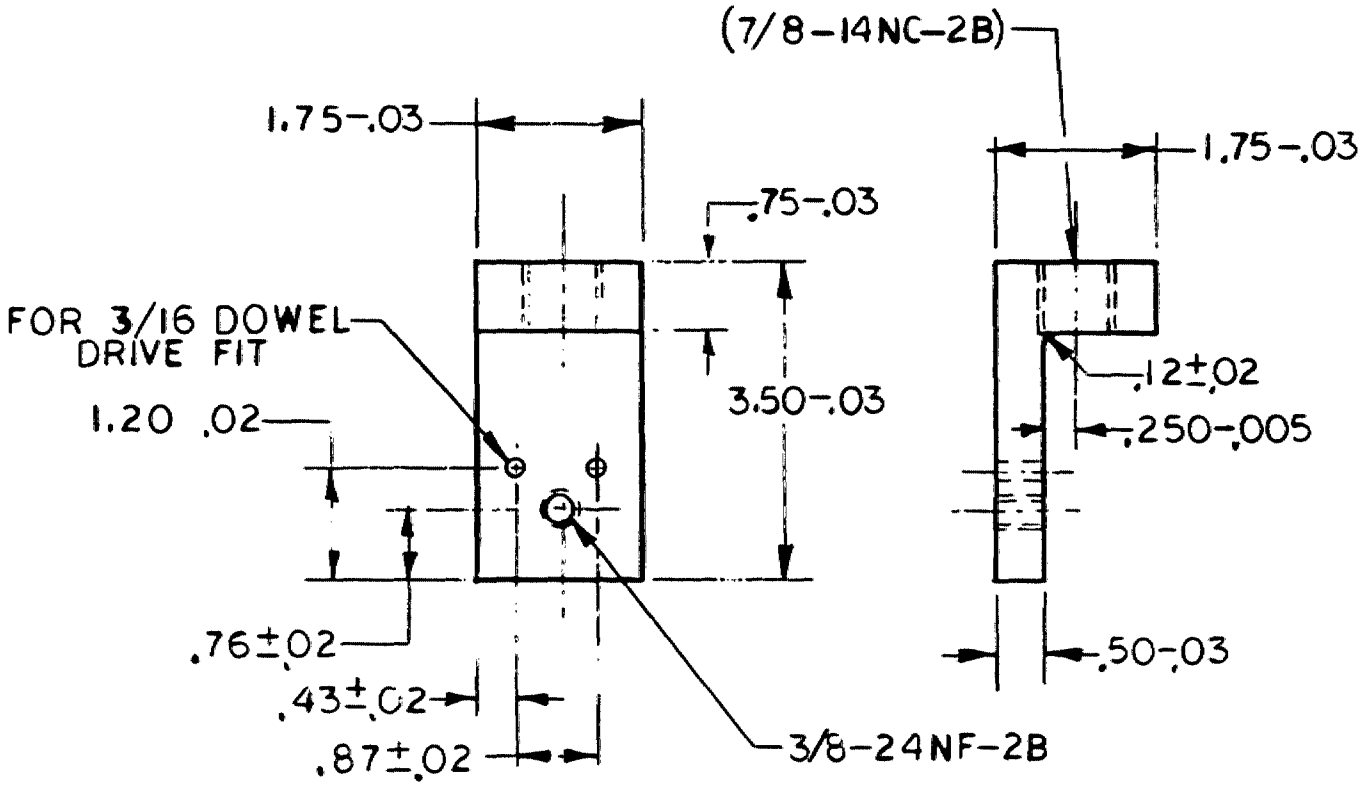


FIGURE 1.D SEAL SPLIT DISC ASSEMBLY

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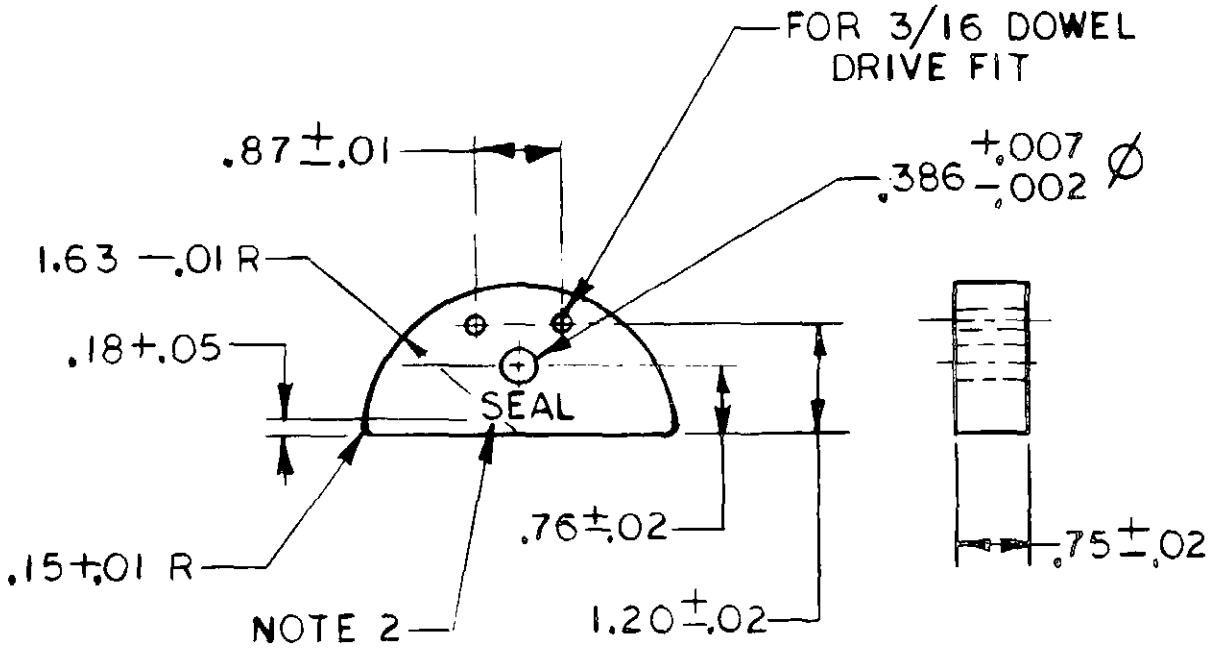


NOTES:-

1 MATERIAL STEEL, HOT ROLLED ALLOY BARS, GRADE 4340, ASTM-A322, 2 REQ'D.

FIGURE 1.E BASE SPLIT DISC (SEAL)

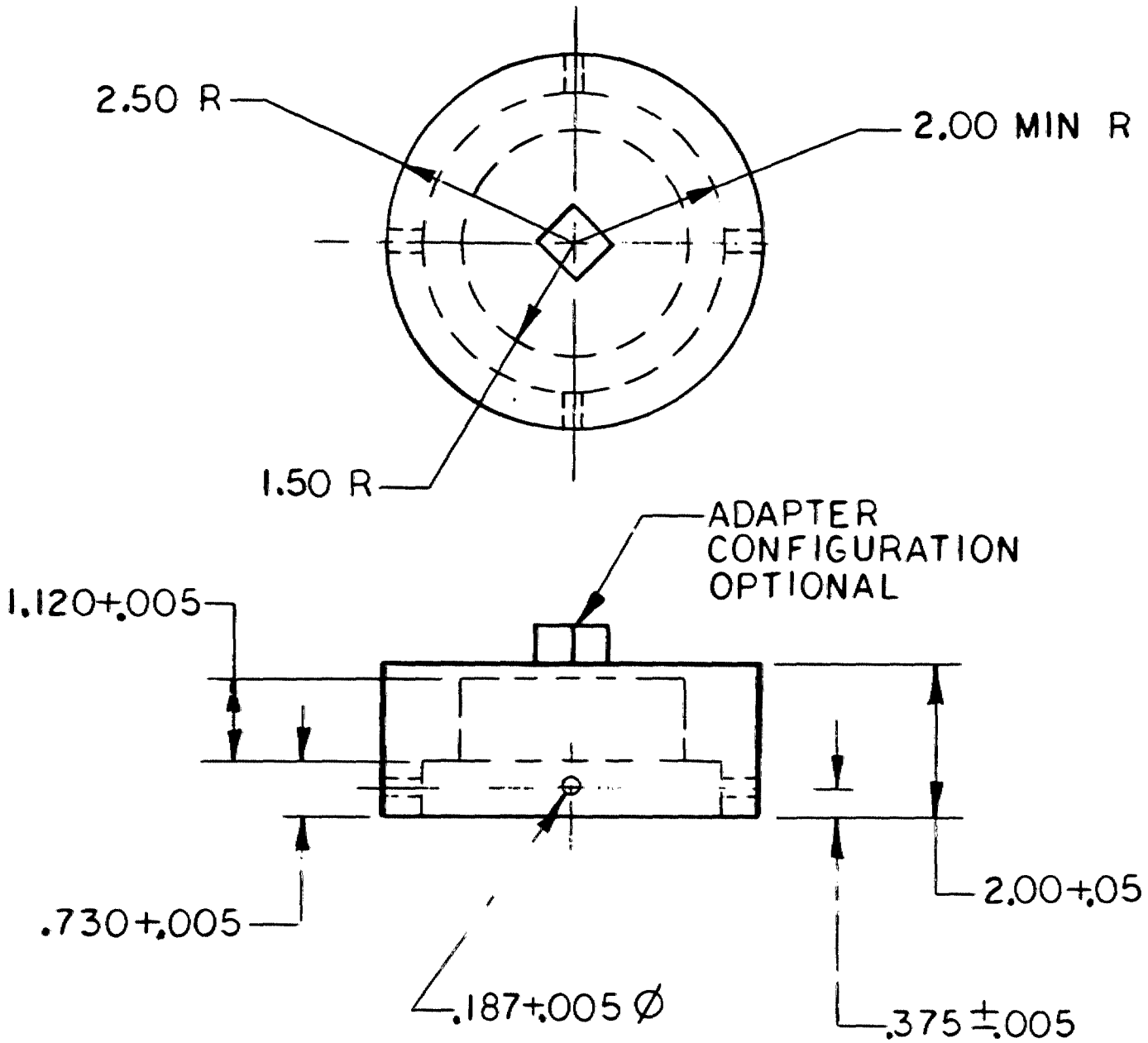
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NOTES:-
1-MATERIAL STEEL, HOT-ROLLED ALLOY BARS, GRADE 4340, ASTM-A322, 2 REQ'D.
2-STAMP AS SHOWN WITH .25-.06 HIGH LETTERS.

FIGURE 1.F SPLIT DISC (SEAL)

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TORQUE FIXTURE
FIGURE 2

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Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1315-A569)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-60646B (AR)	PROJECTILE, 105MM, HEAT-T, M456A1 METAL PARTS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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